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Reducing parasitic capacitance

Abstract

An apparatus is disclosed for reducing parasitic capacitance. In an example aspect, an apparatus includes an amplifier having a differential cascode configuration. Each stack of the amplifier includes a first transistor configured to operate as an input stage and a second transistor configured to operate as a cascode stage. The first and second transistors each include two channel terminal regions having a doping type that is uniform across the two channel terminal regions. Surfaces of first channel terminal regions of the first and second transistors abut a first and second quantity of electrical contacts, respectively. Second channel terminal regions of the first and second transistors form a floating region at a floating node. Each of the first quantity of electrical contacts and the second quantity of electrical contacts is greater than a third quantity of electrical contacts abutting a surface of the floating region.

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Background/Summary

TECHNICAL FIELD

(1) This disclosure relates generally to wireless transceivers and other components that employ amplifiers and, more specifically, to reducing parasitic capacitance within an amplifier.

BACKGROUND

(2) To increase transmission rates and throughput, cellular and other wireless networks are using signals with higher frequencies and smaller wavelengths. As an example, 5.sup.th generation (5G)-capable devices or next-generation wireless local area network (WLAN)-capable devices communicate with networks using frequencies that include those at or near the extremely high frequency (EHF) spectrum (e.g., frequencies greater than 24 gigahertz (GHz)) with wavelengths at or near millimeter wavelengths (mmW). However, these signals present various technological challenges, such as higher path loss as compared to signals for earlier generations of wireless communications. In certain scenarios, it can be difficult for a mmW wireless signal to travel far enough to make cellular or WLAN communications feasible at these higher frequencies.

SUMMARY

(3) An apparatus is disclosed that reduces parasitic capacitance. In example aspects, an amplifier with a differential cascode configuration has an input stage and a cascode stage, which are coupled together at a floating node. Channel terminal regions of the input stage and the cascode stage form a floating region at the floating node. To manage the parasitic capacitance between a gate terminal of the input stage and the floating region, few (if any) electrical contacts are deposited on a surface of the floating region. As such, a smaller quantity of electrical contacts abut the floating region compared to other channel terminal regions of the input stage and the cascode stage. By having fewer electrical contacts within the floating region, the parasitic capacitance of the amplifier can be smaller than other amplifier designs with larger quantities of electrical contacts within the floating region. Also, a footprint of the amplifier can be smaller than other amplifier designs that use neutralizing capacitors.

(4) In an example aspect, an apparatus for amplification is disclosed. The apparatus includes an amplifier having a differential cascode configuration. The amplifier includes at least two stacks. Each stack of the at least two stacks includes a first transistor and a second transistor. The first transistor is configured to operate as an input stage of the amplifier. The second transistor is coupled to the first transistor at a floating node and is configured to operate as a cascode stage of the amplifier. The first and second transistors each include two channel terminal regions having a doping type that is uniform across the two channel terminal regions. The two channel terminal regions include a first channel terminal region and a second channel terminal region. A surface of the first channel terminal region of the first transistor abuts a first quantity of electrical contacts. A surface of the first channel terminal region of the second transistor abuts a second quantity of electrical contacts. The second channel terminal regions of the first and second transistors form a floating region at the floating node. Each of the first quantity of electrical contacts and the second quantity of electrical contacts is greater than a third quantity of electrical contacts abutting a surface of the floating region.

(5) In an example aspect, an apparatus for amplification is disclosed. The apparatus includes means for amplifying a differential signal using a differential cascode configuration. The means for amplifying includes an input stage and a cascode stage, which is coupled to the input stage at a floating node. The input stage and the cascode stage each include two channel terminal regions having a doping type that is uniform across the two channel terminal regions. One of the two channel terminal regions of the input stage and one of the two channel terminal regions of the cascode stage forming a floating region at the floating node. The apparatus also includes means for setting a parasitic capacitance between a gate terminal of the input stage and the one of the two channel terminal regions of the input stage based on a quantity of electrical contacts that abut a surface of the floating region.

(6) In an example aspect, an apparatus for amplification is disclosed. The apparatus includes an amplifier having a differential cascode configuration. The amplifier includes at least two stacks. Each stack of the at least two stacks includes a first transistor and a second transistor. The first transistor is configured to operate as an input stage of the amplifier. The second transistor is coupled to the first transistor at a floating node and is configured to operate as a cascode stage of the amplifier. The first transistor and the second transistor each include two channel terminal regions having a doping type that is uniform across the two channel terminal regions. The two channel terminal regions include a first channel terminal region and a second channel terminal region. The second channel terminal regions of the first and second transistors form a floating region at the floating node. The floating region includes an electrical-contact-free region.

(7) In an example aspect, an apparatus is disclosed. The apparatus includes an amplifier having a differential cascode configuration. The amplifier includes at least two stacks. Each stack of the at least two stacks includes a first transistor and a second transistor. The first transistor is configured to operate as an input stage of the amplifier. The second transistor is coupled to the first transistor at a floating node and is configured to operate as a cascode stage of the amplifier. The amplifier also includes a first capacitor and a second capacitor. The first capacitor is coupled between a gate of the first transistor of a first stack of the at least two stacks and the floating node of a second stack of the at least two stacks. The second capacitor is coupled between a gate of the first transistor of the second stack and the floating region of the first stack. The first transistor and the second transistor each include two channel terminal regions having a doping type that is uniform across the two channel terminal regions. The two channel terminal regions include a first channel terminal region and a second channel terminal region. Surfaces of the second channel terminal regions of the first and second transistors form a floating region at the floating node. Each of a quantity of electrical contacts abutting the first channel terminal region of the first transistor and a quantity of electrical contacts abutting the first channel terminal region of the second transistor is greater than a quantity of electrical contacts abutting a surface of the floating region.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 illustrates an example operating environment for reducing parasitic capacitance.
- (2) FIG. 2 illustrates an example wireless transceiver including at least one amplifier that reduces parasitic capacitance.
- (3) FIG. 3 illustrates example components of an amplifier that reduces parasitic capacitance.
- (4) FIG. 4 illustrates example components of a transistor.
- (5) FIG. 5 illustrates an example implementation of an amplifier that reduces parasitic capacitance.
- (6) FIG. 6 illustrates another example implementation of an amplifier that reduces parasitic capacitance.
- (7) FIG. 7 illustrates an example layout of one of the stacks of an amplifier that reduces parasitic capacitance.
- (8) FIG. 8 illustrates yet another example implementation of an amplifier that reduces parasitic capacitance.
- (9) FIG. 9 illustrates an example layout of an amplifier that reduces parasitic capacitance.
- (10) FIG. 10 illustrates an example vertical layout of an amplifier that reduces parasitic capacitance.

DETAILED DESCRIPTION

(11) Cellular and other wireless networks can use signals with higher frequencies and smaller wavelengths to increase transmission rates and throughput. Signals within the extremely high frequency (EHF) spectrum (e.g., frequencies greater than 24 gigahertz (GHz)) with wavelengths at

or near millimeter wavelengths, however, experience higher path loss compared to signals at lower frequency ranges. As such, it can be difficult for a mmW wireless signal to travel far enough to make cellular or WLAN communications feasible at these higher frequencies.

(12) Electronic devices can employ amplifiers to amplify signals that are transmitted or received. Performance of an amplifier depends on several factors, including an amount of parasitic capacitance that is present within the amplifier. Parasitic capacitance refers to an unwanted capacitive effect that forms between two parts that are in close proximity to one another and have different voltages. Within the amplifier, the parasitic capacitance can degrade output power, gain, efficiency, linearity, and/or stability, particularly at frequencies within the extremely high frequency spectrum.

(13) Some techniques connect neutralizing capacitors to transistors within an amplifier to reduce the amount of parasitic capacitance associated with the transistors. The neutralizing capacitors, however, can increase a footprint of the amplifier. It can be challenging to reduce parasitic capacitance while keeping a footprint of the amplifier sufficiently small to fit within space-constrained devices, such as portable electronic devices like smartphones or wearable devices. Further, amplifiers with larger footprints result in larger integrated circuits that cost more.

(14) To address this challenge, example techniques for reducing parasitic capacitance are described. In example aspects, an amplifier with a differential cascode configuration has an input stage and a cascode stage, which are coupled together at a floating node. Channel terminal regions of the input stage and the cascode stage form a floating region at the floating node. To manage the parasitic capacitance between a gate terminal of the input stage and the floating region, few (if any) electrical contacts are deposited on a surface of the floating region. As such, a smaller quantity of electrical contacts abut the floating region compared to other channel terminal regions of the input stage and the cascode stage. By having fewer electrical contacts within the floating region, the parasitic capacitance of the amplifier can be smaller than other amplifier designs with larger quantities of electrical contacts within the floating region. Also, a footprint of the amplifier can be smaller than other amplifier designs that use neutralizing capacitors.

(15) FIG. 1 illustrates an example environment **100** for operating an amplifier that reduces parasitic capacitance. In the environment **100**, a computing device **102** communicates with a base station **104** through a wireless communication link **106** (wireless link **106**). In this example, the computing device **102** is depicted as a smartphone. However, the computing device **102** can be implemented as any suitable computing or electronic device, such as a modem, a cellular base station, a broadband router, an access point, a cellular phone, a gaming device, a navigation device, a media device, a laptop computer, a desktop computer, a tablet computer, a wearable computer, a server, a network-attached storage (NAS) device, a smart appliance or other internet of things (IoT) device, a medical device, a vehicle-based communication system, a radar, a radio apparatus, and so forth.

(16) The base station **104** communicates with the computing device **102** via the wireless link **106**, which can be implemented as any suitable type of wireless link. Although depicted as a tower of a cellular network, the base station **104** can represent or be implemented as another device, such as a satellite, a server device, a terrestrial television broadcast tower, an access point, a peer-to-peer device, a mesh network node, and so forth. Therefore, the computing device **102** may communicate with the base station **104** or another device via a wireless connection.

(17) The wireless link **106** can include a downlink of data or control information communicated from the base station **104** to the computing device **102**, an uplink of other data or control information communicated from the computing device **102** to the base station **104**, or both a downlink and an uplink. The wireless link **106** can be implemented using any suitable communication protocol or standard, such as 2.sup.nd-generation (2G), 3.sup.rd-generation (3G), 4.sup.th-generation (4G), or 5.sup.th-generation (5G) cellular; IEEE 802.11 (e.g., Wi-Fi®); IEEE 802.15 (e.g., Bluetooth®); IEEE 802.16 (e.g., WiMAX®); and so forth. In some implementations, the wireless link **106** may wirelessly provide power and the base station **104** or the computing

device **102** may comprise a power source.

(18) As shown, the computing device **102** includes an application processor **108** and a computer-readable storage medium **110** (CRM **110**). The application processor **108** can include any type of processor, such as a multi-core processor, that executes processor-executable code stored by the CRM **110**. The CRM **110** can include any suitable type of data storage media, such as volatile memory (e.g., random access memory (RAM)), non-volatile memory (e.g., Flash memory), optical media, magnetic media (e.g., disk), and so forth. In the context of this disclosure, the CRM **110** is implemented to store instructions **112**, data **114**, and other information of the computing device **102**, and thus does not include transitory propagating signals or carrier waves.

(19) The computing device **102** can also include input/output ports **116** (I/O ports **116**) and a display **118**. The I/O ports **116** enable data exchanges or interaction with other devices, networks, or users. The I/O ports **116** can include serial ports (e.g., universal serial bus (USB) ports), parallel ports, audio ports, infrared (IR) ports, user interface ports such as a touchscreen, and so forth. The display **118** presents graphics of the computing device **102**, such as a user interface associated with an operating system, program, or application. Alternatively or additionally, the display **118** can be implemented as a display port or virtual interface, through which graphical content of the computing device **102** is presented.

(20) A wireless transceiver **120** of the computing device **102** provides connectivity to respective networks and other electronic devices connected therewith. The wireless transceiver **120** can facilitate communication over any suitable type of wireless network, such as a wireless local area network (WLAN), peer-to-peer (P2P) network, mesh network, cellular network, ultra-wideband (UWB) network, wireless wide-area-network (WWAN), and/or wireless personal-area-network (WPAN). In the context of the example environment **100**, the wireless transceiver **120** enables the computing device **102** to communicate with the base station **104** and networks connected therewith. However, the wireless transceiver **120** can also enable the computing device **102** to communicate “directly” with other devices or networks.

(21) The wireless transceiver **120** includes circuitry and logic for transmitting and receiving communication signals via an antenna **122**. Components of the wireless transceiver **120** can include amplifiers, switches, mixers, analog-to-digital converters, filters, and so forth for conditioning the communication signals (e.g., for generating or processing signals). The wireless transceiver **120** can also include logic to perform in-phase/quadrature (I/Q) operations, such as synthesis, encoding, modulation, decoding, demodulation, and so forth. In some cases, components of the wireless transceiver **120** are implemented as separate transmitter and receiver entities. Additionally or alternatively, the wireless transceiver **120** can be realized using multiple or different sections to implement respective transmitting and receiving operations (e.g., separate transmit and receive chains). In general, the wireless transceiver **120** processes data and/or signals associated with communicating data of the computing device **102** over the antenna **122**.

(22) In the example shown in FIG. 1, the wireless transceiver **120** includes at least one amplifier **124**. The amplifier **124** can implement aspects of reducing parasitic capacitance, which refers to the act, condition/state, or configuration of reducing and/or cancelling parasitic capacitance by an appreciable level, such as by a sufficient amount so that a target performance is realized. By reducing parasitic capacitance, for instance, the amplifier **124** can realize a target output power, a target gain, a target efficiency, a target linearity, and/or a target stability for wireless applications, including those that utilize frequencies within the extremely high frequency spectrum. The amplifier **124** is further described with respect to FIG. 2.

(23) FIG. 2 illustrates an example wireless transceiver **120**. In the depicted configuration, the wireless transceiver **120** includes a transmitter **202** and a receiver **204**, which are respectively coupled to a first antenna **122-1** and a second antenna **122-2**. In other implementations, the transmitter **202** and the receiver **204** can be connected to a same antenna through a duplexer (not shown) or some other interface (e.g., for a time-division duplexing (TDD) system such as many

millimeter-wave systems). The transmitter **202** is shown to include at least one digital-to-analog converter **206** (DAC **206**), at least one first mixer **208-1**, and at least one amplifier **124** (e.g., a power amplifier). The receiver **204** includes at least one amplifier **210** (e.g., a low-noise amplifier), at least one second mixer **208-2**, and at least one analog-to-digital converter **212** (ADC **212**). The first mixer **208-1** and the second mixer **208-2** are coupled to a local oscillator **214**. Although not explicitly shown, the digital-to-analog converter **206** of the transmitter **202** and the analog-to-digital converter **212** of the receiver **204** can be coupled to the application processor **108** (of FIG. **1**) or another processor associated with the wireless transceiver **120** (e.g., a modem).

(24) In some implementations, the wireless transceiver **120** is implemented using multiple circuits (e.g., multiple integrated circuits), such as a transceiver circuit **216** and a radio-frequency front-end (RFFE) circuit **218**. As such, the components that form the transmitter **202** and the receiver **204** are distributed across these circuits. As shown in FIG. **2**, the transceiver circuit **216** includes the digital-to-analog converter **206** of the transmitter **202**, the mixer **208-1** of the transmitter **202**, the mixer **208-2** of the receiver **204**, and the analog-to-digital converter **212** of the receiver **204**. In other implementations, the digital-to-analog converter **206** and the analog-to-digital converter **212** can be implemented on another separate circuit that includes the application processor **108** or the modem. The radio-frequency front-end circuit **218** includes the amplifier **124** of the transmitter **202** and the amplifier **210** of the receiver **204**.

(25) During transmission, the transmitter **202** generates a radio-frequency transmit signal **220**, which is transmitted using the antenna **122-1**. To generate the radio-frequency transmit signal **220**, the digital-to-analog converter **206** provides a transmit signal **222** to the first mixer **208-1**. The transmit signal **222** can be a baseband signal or an intermediate-frequency signal. The first mixer **208-1** upconverts the transmit signal **222** using a local oscillator (LO) signal **224** provided by the local oscillator **214**. The first mixer **208-1** generates an upconverted version of the transmit signal **222**, which can be a radio-frequency signal. The amplifier **124** amplifies the upconverted transmit signal **222** and passes an amplified transmit signal **226** to the antenna **122-1** for transmission. The transmitted amplified transmit signal **226** is represented by the radio-frequency transmit signal **220**.

(26) During reception, the antenna **122-2** receives a radio-frequency receive signal **228** and passes the radio-frequency receive signal **228** to the receiver **204**. The amplifier **210** of the receiver **204** amplifies the radio-frequency receive signal **228** and passes an amplified receive signal **230** to the second mixer **208-2**. The second mixer **208-2** downconverts the amplified receive signal **230** using the local oscillator signal **224**. The analog-to-digital converter **212** converts the downconverted amplified receive signal **230** into a digital signal, which can be processed by the application processor **108** or another processor associated with the wireless transceiver **120** (e.g., the modem).

(27) In the depicted configuration, the amplifier **124** implements a power amplifier within the transmitter **202**. The amplifier **124**, however, is not limited to implementations associated with power amplifiers. In general, the amplifier **124** can implement other types of amplifiers within the radio-frequency front-end circuit **218** and/or the transceiver circuit **216**. For instance, the amplifier **124** can implement a low-noise amplifier within the receiver **204**. Additionally or alternatively, the amplifier **124** can implement another amplifier, such as a variable-gain amplifier, within the transmitter **202** or the receiver **204**.

(28) FIG. **2** illustrates one example configuration of the wireless transceiver **120**. Other configurations of the wireless transceiver **120** can support multiple frequency bands and share an antenna **122** across multiple transceivers. One of ordinary skill in the art can appreciate the variety of other configurations for which amplifier **124** may be included. For example, the amplifier **124** can be implemented within a wireless transceiver **120** that has a superheterodyne architecture or a direct-conversion architecture (or a zero-intermediate-frequency architecture). In some aspects, the amplifier **124** can be integrated within active phase shifters of the wireless transceiver **120**. The amplifier **124** can have a differential cascode configuration, which is further described with respect to FIG. **3**.

(29) FIG. 3 illustrates example components of the amplifier **124**, which has a differential cascode configuration **302**. In particular, the amplifier **124** includes at least two stacks (or at least one pair of stacks), which are represented by a first stack **304** and a second stack **306**. Each stack **304** and **306** represents an amplification branch of the amplifier **124**. The two stacks **304** and **306** enable the amplifier **124** to accept a differential input signal (e.g., two complementary input signals or two input signals with opposite polarities) and generate a differential output signal (e.g., two complementary output signals or two output signals with opposite polarities). The differential output signal is amplified relative to the differential input signal.

(30) Each stack **304** and **306** includes an input stage **308** and a cascode stage **310**, which are coupled together at a floating node **312**. The floating node **312** has a floating potential (e.g., a floating voltage) because the floating node **312** is not directly connected to a ground or a fixed voltage, such as a supply voltage. As such, the voltage at the floating node **312** can vary during operation of the amplifier **124**. The floating node **312** can also be referred to as an internal node. As an internal node, the floating node **312** is not connected to any components of the wireless transceiver **120** or voltage sources that are external to the amplifier **124**. In this way, the floating node **312** differs from the input node **502** and the output node **506**, which are connected to other components of the wireless transceiver **120**.

(31) For each stack **304** or **306**, the input stage **308** can be implemented using at least one transistor **314**, and the cascode stage **310** can be implemented using at least one other transistor **316**. In an example implementation, the transistor **314** is configured as a common-source amplifier **318** (CS amplifier **318**), and the transistor **316** is configured as a common-gate amplifier **320** (CG amplifier **320**). The transistors **314** and **316** can be implemented using an n-channel metal-oxide-semiconductor field-effect transistor (NMOSFET), a p-channel metal-oxide-semiconductor field-effect transistor (PMOSFET), an extended-drain NMOSFET (ED-NMOSFET), a super-low threshold-voltage NMOSFET (SLVT-NMOSFET), a bipolar junction transistor (BJT), another type of transistor, or combinations thereof.

(32) In an example implementation, the transistors **314** and **316** within each stack **304** and **306** are symmetric **322**. This means the transistor **314** of the input stage **308** and the transistor **316** of the cascode stage **310** are a same type of transistor. For example, the transistors **314** and **316** can both be implemented using NMOSFETs. In another example implementation, the transistors **314** and **316** within each stack **304** and **306** are asymmetric **324**. This means the transistor **314** of the input stage **308** and the transistor **316** of the cascode stage **310** are different types of transistors (e.g., different types of NMOSFETs). For example, the transistor **314** can be implemented using a SLVT-NMOSFET, and the transistor **316** can be implemented using an ED-NMOSFET.

(33) The transistor **314** of the input stage **308** has parasitic capacitance **326**. To at least partially reduce this parasitic capacitance **326**, few (if any) electrical contacts can be disposed on a surface of a floating region associated with the floating node **312**, as further described with respect to FIGS. 4 and 5. Significantly reducing or eliminating the electrical contacts disposed on the surface of the floating region can reduce the parasitic capacitance **326** by up to 60% in some implementations.

(34) Optionally, the amplifier **124** can include two or more capacitors **328** (or neutralization capacitors) to further reduce at least a portion of the parasitic capacitance **326**. The capacitors **328** are coupled between the stacks **304** and **306**, as further described with respect to FIGS. 8 and 9. Capacitances of the capacitors **328** can be appropriately set to at least partially reduce the parasitic capacitance **326**. To avoid increasing a footprint of the amplifier **124**, the capacitors **328** can be integrated within a layout of the stacks **304** and **306**. In other words, the capacitors **328** can be formed using metal interconnect layers associated with the stacks **304** and **306**. By not using distinct, external capacitors, the capacitors **328** can provide additional parasitic-capacitance-reduction capabilities without significantly increasing a footprint of the amplifier **124**. The transistors **314** and **316** are further described with respect to FIG. 4.

(35) FIG. 4 illustrates example components of the transistor **314** or **316**. In the depicted configuration, the transistor **314** or **316** is implemented as a metal-oxide-semiconductor field-effect transistor (MOSFET). Other implementations of the transistors **314** and **316** are also possible. The transistor **314** or **316** includes at least one substrate **402** (or body) with at least two channel terminal regions **404-1** and **404-2**, which are arranged or positioned along an axis **400**. The substrate **402** can be a single-crystal silicon wafer. In general, the substrate **402** provides physical support for components of the amplifier **124**. Sometimes the substrate **402** represents a substrate of an integrated circuit that includes the amplifier **124**. An example integrated circuit can include the radio-frequency front-end circuit **218** and/or the transceiver circuit **216**. The substrate **402** has a first doping type **406-1**.

(36) The two channel terminal regions **404-1** and **404-2** have a second doping type **406-2**, which differs from the first doping type **406-1** of the substrate **402**. Consider an example in which the transistor **314** or **316** represents an NMOSFET. In this case, the doping type **406-1** of the substrate **402** represents a p-type doping, and the doping type **406-2** of the two channel terminal regions **404-1** and **404-2** represent an n-type doping. The doping type (e.g., n-type or p-type doping) within each of the two channel terminal regions **404-1** and **404-2** is uniform. Here, the term “uniform” means that both channel terminal regions **404-1** and **404-2** have a single doping type, such as an n+ doping type, and the profile of the channel terminal regions **404-1** and **404-2** are approximately similar.

(37) The transistor **314** or **316** also includes at least one insulator **408**, which can be grown or disposed on a surface of the substrate **402**. The insulator **408** covers a portion of the surface of the substrate **402** that is between the two channel terminal regions **404-1** and **404-2**. In an example implementation, the insulator **408** is formed using oxide, such as silicon dioxide (SiO₂). A conducting material, such as metal, is deposited on a surface of the insulator **408** to form a gate electrode **410**.

(38) At least one electrical contact **412** is deposited on a surface of at least one of the channel terminal regions **404**, such as the channel terminal region **404-1** in FIG. 4. In general, any quantity of electrical contacts **412** can be deposited on the surface of the channel terminal region **404-1**. For example, the quantity of electrical contacts **412** can be 1, 2, 4, 10, 20, 100, and so forth. An electrical contact **412** may or may not be deposited on a surface of the other channel terminal region **404-2**, as further described below.

(39) The transistor **314** or **316** can be considered a three-terminal device, with the terminals including a gate **414** (e.g., a gate terminal) and two channel terminals **416-1** and **416-2**. The channel terminals **416-1** and **416-2** can each represent a source **418** (e.g., a source terminal) or a drain **420** (e.g., a drain terminal). In example implementations in which the transistor **314** implements the common-source amplifier **318** and the transistor **316** implements the common-gate amplifier **320**, the channel terminal **416-1** represents the source **418** of the transistor **314**, and the channel terminal **416-2** represents the drain **420** of the transistor **314**. For the transistor **316**, the channel terminal **416-1** represents the drain **420**, and the channel terminal **416-2** represents the source **418**.

(40) The transistor **314** can have parasitic capacitance **326**, which exists between the gate **414** and the channel terminal **416-2** (e.g., the drain **420**). This parasitic capacitance **326** is based at least in part on parasitic capacitances of a back-end-of-line (BEOL) of the amplifier **124** and parasitic capacitances between the gate **414** and any electrical contacts **412** disposed on the surface of the channel terminal region **404-2**.

(41) One aspect of reducing the parasitic capacitance **326** reduces the quantity of electrical contacts **412** deposited on the channel terminal region **404-2** such that this quantity is less than a quantity of electrical contacts **412** deposited on the channel terminal region **404-1**. For example, the quantity of electrical contacts **412** deposited on the channel terminal region **404-2** can be between 0%-50% of the quantity of electrical contacts **412** deposited on the channel terminal region **404-1**. In some

implementations, the channel terminal region **404-2** represents an “electrical-contact-free region” in which no electrical contacts are deposited on the surface of the channel terminal region **404-2**. In other implementations, one or more electrical contacts **412** are deposited on the channel terminal region **404-2**. In general, a total surface area of the channel terminal region **404-2** that abuts the electrical contact(s) **412** is less than a total surface area of the channel terminal region **404-1** that abuts the electrical contact(s) **412**. The channel terminal regions **404-2** of the transistors **314** and **316** form a floating region, which is further described with respect to FIG. 5.

(42) FIG. 5 illustrates an example implementation of the amplifier **124**. In the depicted configuration, the amplifier **124** includes the stacks **304** and **306**. Each stack **304** and **306** includes the transistors **314** and **316**, which are configured as a common-source amplifier **318** and a common-gate amplifier **320**, respectively. In FIG. 5, the transistors **314** and **316** are shown to be implemented as NMOSFETs. Other implementations, however, are also possible. For example, the transistors **314** can be SLVT-NMOSFETs, and the transistors **316** can be ED-NMOSFETs.

(43) The transistors **314** and **316** are coupled together within each stack **304** and **306** at the floating node **312**. In particular, the drain **420** of the transistor **314** is coupled to the source **418** of the transistor **316** at the floating node **312**. The drain **420** of the transistor **314** and the source **418** of the transistor **316** are associated with the channel terminal regions **404-2**. The channel terminal regions **404-2** of the transistors **314** and **316** can form a floating region **514**. In some cases, the floating region **514** represents a shared region, which represents both the drain **420** of the transistor **314** and the source **418** of the transistor **316**. Implementing the floating region **514** as a shared region can conserve space within the amplifier **124**. The floating region **514** represents a conducting metal-silicided diffusion region. In particular, a thin layer of silicide, such as nickel silicide, can be present on a surface of the floating region **514**.

(44) Each stack **304** and **306** also includes an input node **502**, a bias node **504**, an output node **506**, and a ground node **508**. The input node **502** is coupled to the gate **414** of the transistor **314**. The bias node **504** is coupled to the gate **414** of the transistor **316**. The stacks **304** and **306** can include at least one capacitor, which is coupled between the gate **414** of the transistor **316** and the bias node **504**. The output node **506** is coupled to the drain **420** of the transistor **316**. Although not explicitly shown, the output node **506** can also be coupled to a load and a power supply node. The ground node **508** is coupled to the source **418** of the transistor **314**.

(45) During operation, the input node **502** accepts a differential input signal (e.g., the transmit signal **222**). The output node **506** provides a differential output signal (e.g., the amplified transmit signal **226**), which is amplified relative to the differential input signal. The bias node **504** is coupled to a supply voltage **510** and provides a bias voltage to the gate **414** of the transistor **316**. The ground node **508** couples the source **418** of the transistor **314** to a ground **512**. Although the amplifier **124** is shown to include one pair of stacks in FIG. 5, some implementations of the amplifier **124** can have additional pairs of stacks, as further described with respect to FIG. 6.

(46) FIG. 6 illustrates another example implementation of the amplifier **124**. In the depicted configuration, the amplifier **124** includes multiple pairs of stacks. In the depicted configuration, the amplifier **124** includes stacks **304-1**, **304-2** . . . **304-N**, where N represents a positive integer. The amplifier **124** also includes stacks **306-1**, **306-2** . . . **306-N**. Each pair of stacks includes one of the stacks **304-1** to **304-N** and one of the stacks **306-1** to **306-N**. For example, a first pair of stacks includes stacks **304-1** and **306-1**, a second pair of stacks includes stacks **304-2** and **306-2**, and an Nth pair of stacks includes stacks **304-N** and **306-N**.

(47) The stacks **304-1** to **304-N** are coupled together in parallel, and the stacks **306-1** to **306-N** are coupled together in parallel. In particular, the stacks **304-1** to **304-N** are coupled together at the input node **502** (not shown), the bias node **504**, the output node **506**, and the ground node **508**. Also, the stacks **306-1** to **306-N** are coupled together at the input node **502** (not shown), the bias node **504**, the output node **506**, and the ground node **508**. To reduce the parasitic capacitance **326**, a quantity of electrical contacts **412** associated with the floating region **514** of the amplifier **124** is

limited, as further described with respect to FIG. 7.

(48) FIG. 7 illustrates an example layout of the stack **304** or **306**. In the depicted configuration, the transistors **314** and **316** are implemented as multi-finger transistors. With this multi-finger design, multiple pairs of parallel stacks can be implemented in a manner that conserves space. The channel terminals **416-1** and **416-2** and gates **414** of the transistors **314** and **316** are arranged or positioned along the axis **400**. Starting from the left side of FIG. 7 and moving to the right along the axis **400**, the stack **304** or **306** includes a first region **702**, which is a channel terminal region **404-1**. Disposed on a surface of the first region **702** are electrical contacts **412**, which connect the first region **702** to at least one metal layer of the transistor **314**. The first region **702** and the electrical contacts **412** form, at least in part, the source **418** of the transistor **314**.

(49) To the right of the source **418** of the transistor **314** is the gate **414** of the transistor **314**. A second region **704** to the right of the gate **414** of the transistor **314** represents the floating region **514** (e.g., the drain **420** of the transistor **314** and the source **418** of the transistor **316**). The floating region **514** is positioned between the gate **414** of the transistor **314** and the gate **414** of the transistor **316**. The floating region **514** can be formed with a single type of doping, such as an n+ doping type.

(50) A third region **706** represents another channel terminal region **404-1**. In particular, the third region **706** corresponds to the drain **420** of the transistor **316**. Disposed on a surface of the third region **706** are electrical contacts **412**, which connect the third region **706** to at least one metal layer of the transistor **316**. The third region **706** and the electrical contacts **412** form, at least in part, the drain **420** of the transistor **316**.

(51) In some implementations, widths of the gates **414** are different between the transistors **314** and **316**. Consider an axis **708**, which is approximately perpendicular to the axis **400**. Along the axis **708**, the fingers associated with the gate **414** of the transistor **314** have a width **710**. Also, the fingers associated with the gate **414** of the transistor **316** have a width **712**. As shown in FIG. 7, the width **712** is greater than the width **710**. The larger width **712** can enhance stability performance of the amplifier **124** by effectively adding resistance between the gate **414** of the transistor **316** and the bias node **504**.

(52) Consider a case in which FIG. 7 illustrates the stack **304**. Although not explicitly shown, a layout of the stack **306** can be positioned to the right or left of the stack **304** along the axis **400**. In some implementations, the source **418** of the transistor **314** of the stack **304** can be extended along the axis **400** and connect to the corresponding source **418** of the transistor **314** of the stack **306**.

(53) To reduce the parasitic capacitance **326**, fewer electrical contacts **412** are disposed on the floating region **514** relative to the other channel terminal regions **404-1**. As an example, the quantity of electrical contacts **412** disposed on the floating region **514** can be 0, 1, 2, 4, 6, or 10 while the quantity of electrical contacts **412** disposed on one of the other channel terminal regions **404-1** is greater. As such, a total surface area of the floating region **514** that abuts the electrical contacts **412** is less than a total surface area of one of the channel terminal regions **404-1** that abuts the electrical contacts **412**. In this example, the floating regions **514** (e.g., the second region **704**) are electrical-contact-free regions in which no electrical contacts are disposed on the surface of the floating regions **514**. In other words, the floating regions **514** lack or are devoid of electrical contacts **412**. Sometimes, this aspect of reducing parasitic capacitance can be sufficient for enabling the amplifier **124** to realize a desired level of performance without the need for neutralization capacitors. If additional parasitic capacitance reduction is required to achieve the desired level of performance, at least one electrical contact **412** can be deposited on each of the floating regions **514** to enable the use of the capacitors **328**, as further described with respect to FIGS. 8 and 9.

(54) FIG. 8 illustrates yet another example implementation of the amplifier **124**. The amplifier **124** of FIG. 8 is similar to the amplifier **124** of FIG. 5, with the addition of two capacitors **328-1** and **328-2**. In this case, at least one electrical contact is disposed on the surface of each of the floating

regions **514** of the stacks **304** and **306**. In the depicted configuration, the capacitor **328-1** is coupled between the at least one electrical contact disposed on the surface of the floating region **514** of the stack **304** and the gate **414** of the common-source amplifier **318** of the stack **306**. Similarly, the capacitor **328-2** is coupled between the at least one electrical contact disposed on the surface of the floating region **514** of the stack **306** and the gate **414** of the common-source amplifier **318** of the stack **304**. In this way, the capacitors **328-1** and **328-2** can further reduce the parasitic capacitance **326** within the common-source amplifiers **318**. To avoid increasing a footprint of the amplifier **124**, the capacitors **328-1** and **328-2** can be integrated within the layout of the stacks **304** and **306**, as further described with respect to FIGS. **9** and **10**.

(55) FIG. **9** illustrates an example layout of the amplifier **124**, with the stack **304** shown on the left and the stack **306** shown on the right. The layout of each stack **304** and **306** in FIG. **9** is similar to the layout shown in FIG. **7**, however the stacks **304** and **306** in FIG. **9** are shown with fewer fingers for simplicity. The capacitors **328-1** and **328-2** can be integrated within a footprint of the amplifier **124**. In this case, the capacitors **328-1** and **328-2** are implemented, at least in part, using the existing back-end-of-line of the amplifier **124**. In this manner, the capacitors **328-1** and **328-2** do not appreciably increase a footprint of the amplifier **124**. Also, because the floating regions **514** (e.g., regions **704**) have fewer electrical contacts **412** than the other channel terminal regions **404-1** (e.g., regions **702** and **706**), capacitances of the capacitors **328-1** and **328-2** can also be smaller compared to the capacitance of neutralization capacitors of other amplifiers that have a greater quantity of electrical contacts **412** disposed on the floating regions **514**. With the described techniques, the parasitic capacitance **326** within the amplifier **124** can be meaningfully reduced without substantially increasing a footprint of the amplifier **124**. The capacitors **328-1** and **328-2** are further described with respect to FIG. **10**.

(56) FIG. **10** illustrates portions of the stacks **304** and **306** as implemented using multiple metal layers **1002** and at least one polysilicon layer **1004** disposed “above” the substrate **402** along an axis **1006**. The axis **1006** is orthogonal to the axes **400** and **708**. Vias **1008** connect the metal layers **1002** and the polysilicon layer **1004** together. For example, a first set of vias **1008** connect the polysilicon layer **1004** and a first set of metal layers **1002** together to form the gate **414**. A second set of vias **1008** connect a second set of metal layers **1002** together to form the floating node **312**. Although five metal layers **1002** are explicitly shown in FIG. **10**, the stacks **304** and **306** can be implemented with any quantity of metal layers **1002**.

(57) In the depicted configuration, the capacitors **328-1** and **328-2** are formed by overlapping one of the metal layers **1002** associated with the appropriate floating node **312** and one of the metal layers **1002** associated with the appropriate gate **414** along the axis **1006**. In particular, the capacitor **328-1** is formed by overlapping a metal layer **1002** associated with the floating node **312** of the stack **304** with a metal layer **1002** associated with the gate **414** of the stack **306**. Also, the capacitor **328-2** is formed by overlapping a metal layer **1002** associated with the floating node **312** of the stack **306** with a metal layer **1002** associated with the gate **414** of the stack **304**. In this way, the capacitors **328-1** and **328-2** are integrated within a layout of the first and second stacks **304** and **306** using the metal layers **1002**. Other possible implementations of the capacitors **328-1** and **328-2** using the metal layers **1002** are also possible. By using the back-end-of-line of the amplifier **124**, the capacitors **328-1** and **328-2** can be integrated without increasing a footprint of the amplifier **124** along the axes **400** and **708**.

(58) Some aspects are described below. Aspect 1: An apparatus comprising: an amplifier having a differential cascode configuration, the amplifier comprising at least two stacks, each stack of the at least two stacks comprising: a first transistor configured to operate as an input stage of the amplifier; and a second transistor coupled to the first transistor at a floating node, the second transistor configured to operate as a cascode stage of the amplifier, wherein: the first and second transistors each comprise two channel terminal regions having a doping type that is uniform across the two channel terminal regions, the two channel terminal regions comprising a first channel

terminal region and a second channel terminal region; a surface of the first channel terminal region of the first transistor abuts a first quantity of electrical contacts; a surface of the first channel terminal region of the second transistor abuts a second quantity of electrical contacts; the second channel terminal regions of the first and second transistors form a floating region at the floating node; and each of the first quantity of electrical contacts and the second quantity of electrical contacts is greater than a third quantity of electrical contacts abutting a surface of the floating region. Aspect 2: The apparatus of aspect 1, wherein: a total surface area of the floating region that abuts the third quantity of electrical contacts is less than a total surface area of the first channel terminal region of the first transistor that abuts the first quantity of electrical contacts; and the total surface area of the floating region that abuts the third quantity of electrical contacts is less than a total surface area of the first channel terminal region of the second transistor that abuts the second quantity of electrical contacts. Aspect 3: The apparatus of aspect 1 or 2, wherein the third quantity of contacts is equal to zero. Aspect 4: The apparatus of aspect 3, wherein: each surface of the first channel terminal regions of the first and second transistors abuts at least one electrical contact; and the surface of the floating region does not abut an electrical contact. Aspect 5: The apparatus of any previous aspect, wherein: the third quantity of electrical contacts is greater than zero; the at least two stacks comprise a first stack and a second stack; the surface of the floating region of the first stack abuts at least one first electrical contact; the surface of the floating region of the second stack abuts at least one second electrical contact; and the amplifier comprises: a first capacitor coupled between the at least one first electrical contact and a gate terminal of the first transistor of the second stack; and a second capacitor coupled between the at least one second electrical contact and a gate terminal of the first transistor of the first stack. Aspect 6: The apparatus of aspect 5, wherein the first and second capacitors are integrated within a layout of the first and second stacks. Aspect 7: The apparatus of any previous aspect, wherein: the first transistor comprises a common-source amplifier; and the second transistor comprises a common-gate amplifier. Aspect 8: The apparatus of any previous aspect, wherein the first transistor and the second transistor are symmetrical for each stack of the at least two stacks. Aspect 9: The apparatus of any one of aspects 1-7, wherein the first transistor and the second transistor are asymmetrical for each stack of the at least two stacks. Aspect 10: The apparatus of aspect 9, wherein: the first transistor comprises an extended-drain n-channel metal-oxide-semiconductor field-effect transistor; and the second transistor comprises a super-low threshold-voltage n-channel metal-oxide-semiconductor field-effect transistor. Aspect 11: The apparatus of any previous aspect, wherein: the amplifier comprises a substrate; the two channel terminal regions of the first and second transistors are positioned within the substrate along a first axis; the first and second transistors each comprise at least one metal layer that forms one or more fingers associated with a gate terminal, the one or more fingers having a width along a second axis that is substantially perpendicular to the first axis; and the width of the one or more fingers of the second transistor is greater than the width of the one or more fingers of the first transistor. Aspect 12: The apparatus of any previous aspect, wherein the floating region comprises a silicided diffusion region positioned between a gate of the first transistor and a gate of the second transistor. Aspect 13: The apparatus of any previous aspect, wherein the at least two stacks comprise multiple pairs of stacks coupled together in parallel. Aspect 14: The apparatus of any previous aspect, further comprising: a wireless transceiver coupled to at least one antenna, the wireless transceiver comprising the amplifier and configured to amplify, using the amplifier, a wireless signal communicated via the at least one antenna. Aspect 15: The apparatus of aspect 14, wherein the amplifier comprises a power amplifier. Aspect 16: An apparatus comprising: means for amplifying a differential signal using a differential cascode configuration, the means for amplifying comprising: an input stage; and a cascode stage coupled to the input stage at a floating node, the input stage and the cascode stage each comprising two channel terminal regions having a doping type that is uniform across the two channel terminal regions, one of the two channel terminal regions of the input stage and one of the two channel terminal regions of the cascode stage forming

a floating region at the floating node; and means for setting a parasitic capacitance between a gate terminal of the input stage and the one of the two channel terminal regions of the input stage based on a quantity of electrical contacts that abut a surface of the floating region. Aspect 17: The apparatus of aspect 16, wherein: another one of the two channel terminal regions of the input stage abuts a first quantity of electrical contacts; another one of the two channel terminal regions of the cascode stage abuts a second quantity of electrical contacts; and each of the first quantity of electrical contacts and the second quantity of electrical contacts is greater than a third quantity of electrical contacts abutting the surface of the floating region. Aspect 18: The apparatus of aspect 17, wherein the quantity of electrical contacts that abut the surface of the floating region is equal to zero. Aspect 19: The apparatus of aspect 17, wherein: the quantity of electrical contacts that abut the surface of the floating region is greater than zero; and the means for setting comprises capacitive means for reducing at least a portion of the parasitic capacitance, the capacitive means coupled to the floating region via electrical contacts. Aspect 20: The apparatus of any one of aspects 16-19, wherein: the input stage and the cascode stage form a stack of the means for amplifying; and the input stage and the cascode stage of the stack are symmetrical. Aspect 21: The apparatus of any one of aspects 16-19, wherein: the input stage and the cascode stage form a stack of the means for amplifying; and the input stage and the cascode stage of the stack are asymmetrical. Aspect 22: An apparatus comprising: an amplifier having a differential cascode configuration, the amplifier comprising at least two stacks, each stack of the at least two stacks comprising: a first transistor configured to operate as an input stage of the amplifier; and a second transistor coupled to the first transistor at a floating node, the second transistor configured to operate as a cascode stage of the amplifier, wherein: the first transistor and the second transistor each comprise two channel terminal regions having a doping type that is uniform across the two channel terminal regions, the two channel terminal regions comprising a first channel terminal region and a second channel terminal region; the second channel terminal regions of the first and second transistors form a floating region at the floating node; and the floating region comprises an electrical-contact-free region. Aspect 23: The apparatus of aspect 22, wherein: each surface of the first channel terminal regions of the first and second transistors abuts at least one electrical contact; and a surface of the floating region does not abut an electrical contact. Aspect 24: The apparatus of aspect 22 or 23, wherein: the first transistor comprises a common-source amplifier; and the second transistor comprises a common-gate amplifier. Aspect 25: The apparatus of any one of aspects 22-24, wherein the first transistor and the second transistor are symmetrical for each stack of the at least two stacks. Aspect 26: The apparatus of any one of aspects 22-24, wherein the first transistor and the second transistor are asymmetrical for each stack of the at least two stacks. Aspect 27: The apparatus of any one of aspects 22-26, wherein: the amplifier comprises a substrate; the two channel terminal regions of the first and second transistors are positioned within the substrate along a first axis; the first and second transistors each comprise at least one metal layer that forms one or more fingers associated with a gate terminal, the one or more fingers having a width along a second axis that is substantially perpendicular to the first axis; and the width of the one or more fingers of the second transistor is greater than the width of the one or more fingers of the first transistor. Aspect 28: An apparatus comprising: an amplifier having a differential cascode configuration, the amplifier comprising at least two stacks, each stack of the at least two stacks comprising: a first transistor configured to operate as an input stage of the amplifier; and a second transistor coupled to the first transistor at a floating node, the second transistor configured to operate as a cascode stage of the amplifier; a first capacitor coupled between a gate of the first transistor of a first stack of the at least two stacks and the floating node of a second stack of the at least two stacks; and a second capacitor coupled between a gate of the first transistor of the second stack and the floating node of the first stack, wherein: the first transistor and the second transistor each comprise two channel terminal regions having a doping type that is uniform across the two channel terminal regions, the two channel terminal regions comprising a first channel terminal region and a second

channel terminal region; surfaces of the second channel terminal regions of the first and second transistors form a floating region at the floating node; and each of a quantity of electrical contacts abutting the first channel terminal region of the first transistor and a quantity of electrical contacts abutting the first channel terminal region of the second transistor is greater than a quantity of electrical contacts abutting a surface of the floating region. Aspect 29: The apparatus of aspect 28, wherein the first and second capacitors are integrated within a layout of the first and second stacks using metal interconnect layers associated with the first and second stacks. Aspect 30: The apparatus of aspect 28 or 29, wherein: the amplifier comprises a substrate; the two channel terminal regions of the first and second transistors are positioned within the substrate along a first axis; the first and second transistors each comprise at least one metal layer that forms one or more fingers associated with a gate terminal, the one or more fingers having a width along a second axis that is substantially perpendicular to the first axis; and the width of the one or more fingers of the second transistor is greater than the width of the one or more fingers of the first transistor.

(59) Unless context dictates otherwise, use herein of the word “or” may be considered use of an “inclusive or,” or a term that permits inclusion or application of one or more items that are linked by the word “or” (e.g., a phrase “A or B” may be interpreted as permitting just “A,” as permitting just “B,” or as permitting both “A” and “B”). As used herein, a phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover: a, b, c, a-b, a-c, b-c, and a-b-c, as well as any combination with multiples of the same element (e.g., a-a, a-a-a, a-a-b, a-a-c, a-b-b, a-c-c, b-b, b-b-b, b-b-c, c-c, and c-c-c or any other ordering of a, b, and c). Further, items represented in the accompanying figures and terms discussed herein may be indicative of one or more items or terms, and thus reference may be made interchangeably to single or plural forms of the items and terms in this written description. Finally, although subject matter has been described in language specific to structural features or methodological operations, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or operations described above, including not necessarily being limited to the organizations in which features are arranged or the orders in which operations are performed.

Claims

1. An apparatus comprising: an amplifier having a differential cascode configuration, the amplifier comprising at least two stacks, each stack of the at least two stacks comprising: a first transistor configured to operate as an input stage of the amplifier; and a second transistor coupled to the first transistor at a floating node, the second transistor configured to operate as a cascode stage of the amplifier, wherein: the first and second transistors each comprise two channel terminal regions having a doping type that is uniform across the two channel terminal regions, the two channel terminal regions comprising a first channel terminal region and a second channel terminal region; a surface of the first channel terminal region of the first transistor abuts a first quantity of electrical contacts; a surface of the first channel terminal region of the second transistor abuts a second quantity of electrical contacts; the second channel terminal regions of the first and second transistors form a floating region at the floating node; and each of the first quantity of electrical contacts and the second quantity of electrical contacts is greater than a third quantity of electrical contacts abutting a surface of the floating region to reduce parasitic capacitance at the second channel terminal region.

2. The apparatus of claim 1, wherein: a total surface area of the floating region that abuts the third quantity of electrical contacts is less than a total surface area of the first channel terminal region of the first transistor that abuts the first quantity of electrical contacts; and the total surface area of the floating region that abuts the third quantity of electrical contacts is less than a total surface area of the first channel terminal region of the second transistor that abuts the second quantity of electrical

contacts.

3. The apparatus of claim 1, wherein the third quantity of contacts is equal to zero.

4. The apparatus of claim 3, wherein: each surface of the first channel terminal regions of the first and second transistors abuts at least one electrical contact; and the surface of the floating region does not abut an electrical contact.

5. The apparatus of claim 1, wherein: the third quantity of electrical contacts is greater than zero; the at least two stacks comprise a first stack and a second stack; the surface of the floating region of the first stack abuts at least one first electrical contact; the surface of the floating region of the second stack abuts at least one second electrical contact; and the amplifier comprises: a first capacitor coupled between the at least one first electrical contact and a gate terminal of the first transistor of the second stack; and a second capacitor coupled between the at least one second electrical contact and a gate terminal of the first transistor of the first stack.

6. The apparatus of claim 5, wherein the first and second capacitors are integrated within a layout of the first and second stacks.

7. The apparatus of claim 1, wherein: the first transistor comprises a common-source amplifier; and the second transistor comprises a common-gate amplifier.

8. The apparatus of claim 1, wherein the first transistor and the second transistor are symmetrical for each stack of the at least two stacks.

9. The apparatus of claim 1, wherein the first transistor and the second transistor are asymmetrical for each stack of the at least two stacks.

10. The apparatus of claim 9, wherein: the first transistor comprises an extended-drain n-channel metal-oxide-semiconductor field-effect transistor; and the second transistor comprises a super-low threshold-voltage n-channel metal-oxide-semiconductor field-effect transistor.

11. The apparatus of claim 1, wherein: the amplifier comprises a substrate; the two channel terminal regions of the first and second transistors are positioned within the substrate along a first axis; the first and second transistors each comprise at least one metal layer that forms one or more fingers associated with a gate terminal, the one or more fingers having a width along a second axis that is substantially perpendicular to the first axis; and the width of the one or more fingers of the second transistor is greater than the width of the one or more fingers of the first transistor.

12. The apparatus of claim 1, wherein the floating region comprises a silicided diffusion region positioned between a gate of the first transistor and a gate of the second transistor.

13. The apparatus of claim 1, wherein the at least two stacks comprise multiple pairs of stacks coupled together in parallel.

14. The apparatus of claim 1, further comprising: a wireless transceiver coupled to at least one antenna, the wireless transceiver comprising the amplifier and configured to amplify, using the amplifier, a wireless signal communicated via the at least one antenna.

15. The apparatus of claim 14, wherein the amplifier comprises a power amplifier.

16. An apparatus comprising: means for amplifying a differential signal using a differential cascode configuration, the means for amplifying comprising: an input stage; and a cascode stage coupled to the input stage at a floating node, the input stage and the cascode stage each comprising two channel terminal regions having a doping type that is uniform across the two channel terminal regions, the two channel terminal regions comprising a first channel terminal region and a second channel terminal region, the second channel terminal region of the input stage and the second channel terminal region of the cascode stage forming a floating region at the floating node; and means for reducing a parasitic capacitance between a gate terminal of the input stage and the second channel terminal region of the input stage based on a quantity of electrical contacts that abut a surface of the floating region being less than a quantity of electrical contacts that abut a surface of the first channel terminal region.

17. The apparatus of claim 16, wherein the quantity of electrical contacts abutting the surface of the floating region is equal to zero.

18. The apparatus of claim 16, wherein: the quantity of electrical contacts abutting the surface of the floating region is greater than zero; and the means for setting comprises capacitive means for reducing at least a portion of the parasitic capacitance, the capacitive means coupled to the floating region via electrical contacts.
19. The apparatus of claim 16, wherein: the input stage and the cascode stage form a stack of the means for amplifying; and the input stage and the cascode stage of the stack are symmetrical.
20. The apparatus of claim 16, wherein: the input stage and the cascode stage form a stack of the means for amplifying; and the input stage and the cascode stage of the stack are asymmetrical.
21. An apparatus comprising: an amplifier having a differential cascode configuration, the amplifier comprising at least two stacks, each stack of the at least two stacks comprising: a first transistor configured to operate as an input stage of the amplifier; and a second transistor coupled to the first transistor at a floating node, the second transistor configured to operate as a cascode stage of the amplifier; a first capacitor coupled between a gate of the first transistor of a first stack of the at least two stacks and the floating node of a second stack of the at least two stacks; and a second capacitor coupled between a gate of the first transistor of the second stack and the floating node of the first stack, wherein: the first transistor and the second transistor each comprise two channel terminal regions having a doping type that is uniform across the two channel terminal regions, the two channel terminal regions comprising a first channel terminal region and a second channel terminal region; surfaces of the second channel terminal regions of the first and second transistors form a floating region at the floating node; and each of a quantity of electrical contacts abutting the first channel terminal region of the first transistor and a quantity of electrical contacts abutting the first channel terminal region of the second transistor is greater than a quantity of electrical contacts abutting a surface of the floating region to reduce parasitic capacitance at the second channel terminal region.
22. The apparatus of claim 21, wherein the first and second capacitors are integrated within a layout of the first and second stacks using metal interconnect layers associated with the first and second stacks.
23. The apparatus of claim 21, wherein: the amplifier comprises a substrate; the two channel terminal regions of the first and second transistors are positioned within the substrate along a first axis; the first and second transistors each comprise at least one metal layer that forms one or more fingers associated with a gate terminal, the one or more fingers having a width along a second axis that is substantially perpendicular to the first axis; and the width of the one or more fingers of the second transistor is greater than the width of the one or more fingers of the first transistor.
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